

Snug Truncs for Countably Cozero-Presented Compactifications

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Status. This packet gives a candidate substantial partial result toward Ball's Conjecture 6.1.4. The proof is complete for the stated spatial, countably cozero-presented class. It does not settle arbitrary compactifications of Lindelöf frames.

1 The question and the corrected formulation

Let $q : M \rightarrow L$ be a compactification: a dense surjective pointed frame homomorphism with compact regular domain. Ball defines $\mathcal{E}_0 q$ to be the pointed real maps on L that have an almost-finite extended-real lift to M . A trunc $G \subseteq \mathcal{E}_0 q$ is *snug* when its bounded cozero and con sets generate M , and when the image sublocale $q_* L$ is exactly the intersection of the open sublocales determined by the finite domains of the lifts of all $g \in G$ [2, Definition preceding Theorem 3.3.2].

The published conjecture asks whether every compactification $q : M \rightarrow L$ of a Lindelöf frame has such a snug trunc [2, Conjecture 6.1.4]. The arXiv v1 formula accidentally ranges the last intersection over $g \in \overline{G}$, the range of truncation. The published definition corrects this to $g \in G$, in agreement with the proof of the representation theorem. We address the corrected published question.

2 The countable-cozero theorem

We use the spatial notation most natural for the construction. If $(X, *)$ is compact Hausdorff pointed and $Y \subseteq X$ is a dense pointed subspace, restriction gives a compactification

$$q : \mathcal{O}_*(X) \longrightarrow \mathcal{O}_*(Y).$$

For a continuous $\overline{\mathbb{R}}$ -valued function f , write $\text{dom } f = f^{-1}(\mathbb{R})$.

Theorem 1 (countable cozero presentation). *Let $(X, *)$ be compact Hausdorff pointed, let $Y \subseteq X$ be dense with $*$ $\in Y$, and let $q : \mathcal{O}_*(X) \rightarrow \mathcal{O}_*(Y)$ be restriction. Suppose there are cozero opens*

$$X \supseteq U_1 \supseteq U_2 \supseteq \cdots \supseteq Y$$

such that, as sublocales of $\mathcal{O}_(X)$,*

$$q_* \mathcal{O}_*(Y) = \bigcap_{n \geq 1} (U_n \rightarrow \mathcal{O}_*(X)).$$

Then q admits a snugly embedded trunc $G \subseteq \mathcal{E}_0 q$.

The conclusion therefore covers the spatial part of the conjecture whenever the compactification sublocale has a countable cozero-neighborhood presentation. Finite presentations and dense cozero subspaces are included.

3 A hierarchy of pole functions

Choose continuous functions $h_n : X \rightarrow [0, 1]$ such that $U_n = \{h_n > 0\}$. Since $*$ $\in Y$, every $h_n(*) > 0$. Define continuous functions $R_n : X \rightarrow [1, \infty]$ recursively by

$$R_1 = \frac{1}{h_1}, \quad R_{n+1} = \frac{R_n^2}{h_{n+1}}, \quad (1)$$

with the natural value ∞ when a denominator vanishes. The map $(r, t) \mapsto r^2/t$, from $[1, \infty] \times [0, 1]$ to $[1, \infty]$, is continuous with this convention. Induction gives

$$\text{dom } R_n = U_n.$$

The decisive feature is strict asymptotic separation. From

$$\frac{R_n}{R_{n+1}} = \frac{h_{n+1}}{R_n}$$

we see that R_n/R_{n+1} extends continuously by zero on $X \setminus U_{n+1}$. Multiplying these ratios yields

$$\frac{R_i}{R_j} \text{ extends continuously by zero on } X \setminus U_j, \quad i < j. \quad (2)$$

Thus later poles dominate every earlier pole at their common boundary.

Put

$$g_n = (R_n - R_n(*))|_Y$$

and let

$$B = \{b|_Y : b \in C(X, \mathbb{R}), b(*) = 0\}.$$

Finally, let G be the vector sublattice of $C(Y, \mathbb{R})$ generated by $B \cup \{g_n : n \geq 1\}$.

Lemma 1 (finite asymptotic-scale calculus). *Every element of G has a continuous lift $\tilde{g} : X \rightarrow \overline{\mathbb{R}}$, finite on Y , with $\tilde{g}(*) = 0$.*

Proof. First consider a linear form involving only the first N poles,

$$\ell = b + \sum_{i=1}^N a_i g_i, \quad b \in B.$$

If k is the largest index for which $a_k \neq 0$, then (2) gives

$$\frac{\ell}{R_k} \longrightarrow a_k \quad \text{continuously on } X \setminus U_k.$$

Hence ℓ extends there with value $+\infty$ or $-\infty$ according to the sign of a_k ; if every $a_i = 0$, it already extends as the bounded function b .

Every finite lattice-linear expression can, by distributivity, be written as a finite supremum of finite infima of such linear forms. At a boundary point, compare two forms by reading their pole-coefficient vectors in the order

$$(a_N, a_{N-1}, \dots, a_1).$$

The largest index at which the vectors differ determines their order near the point by (2). If all coefficients belonging to poles infinite at that point agree, the common divergent part cancels and only bounded continuous remainders remain. Therefore finite maxima and minima have continuous extended-real limits as well. This supplies the asserted lift of every element of G . All generators vanish at $*$, so every lift does also. $\square$

4 Proof of Theorem 1

For $g \in G$, Lemma 1 gives $\tilde{g} \in \mathcal{D}_0 X$. Restricting its inverse-image frame map to Y gives the element represented by g , so the defining square for $\mathcal{E}_0 q$ commutes. Hence $G \subseteq \mathcal{E}_0 q$.

The set G is already a vector lattice. It is also closed under the truncation operation. Indeed, for $g \geq 0$, compose its lift with the continuous clamp

$$\tau : \overline{\mathbb{R}} \longrightarrow [0, 1], \quad \tau(t) = \min\{\max\{t, 0\}, 1\}.$$

The function $\tau \circ \tilde{g}$ is bounded and continuous on X , and it vanishes at $*$. Consequently $\bar{g} = (\tau \circ \tilde{g})|_Y$ belongs to $B \subseteq G$. Thus G is a trunc in $\mathcal{E}_0 q$.

It remains to verify snugness. First, the cozero and con sets arising from the bounded functions in B generate $\mathcal{O}(X)$. If $x \neq *$ lies in an open V , compact Hausdorff normality gives a function $b : X \rightarrow [0, 1]$, with $b(*) = 0$, such that $x \in \text{coz } b \subseteq V$. If $* \in V$, it gives a function $b : X \rightarrow [0, 1]$, with $b(*) = 0$ and $b = 1$ on $X \setminus V$, so that $* \in \{b < 1\} \subseteq V$. These functions are fixed by truncation, and hence the required $\text{coz } \overline{G}$ and $\text{con } \overline{G}$ sets join-generate $\mathcal{O}(X)$.

Second, the lift of g_n is $R_n - R_n(*)$, so

$$\text{dom } \tilde{g}_n = U_n.$$

Every lift of an element of G is finite on the image sublocale, while the displayed generators already give

$$\bigcap_{g \in G} (\text{dom } \tilde{g} \rightarrow \mathcal{O}_*(X)) \subseteq \bigcap_{n \geq 1} (U_n \rightarrow \mathcal{O}_*(X)) = q_* \mathcal{O}_*(Y).$$

The reverse inclusion follows because every $g \in G$ is finite on Y . The finite-domain intersection is therefore exact. Both snugness conditions hold.

5 Why the unrestricted upgrade remains open

The construction succeeds because every finite expression has a largest pole index. A successor pole can always be made to dominate a previous master scale using (1); countable stages can be arranged in a sequence. For a compactification whose cozero neighborhoods have uncountable cofinality, there need not be a continuous real-valued master pole dominating all earlier directions at a limit stage. Lindelöfness of L guarantees an intersection by cozero sublocales, but it does not give a countable cozero presentation in a fixed compactification. Arbitrarily adjoining reciprocal poles is invalid because differences of two incomparable infinities need not extend.

Bounded searches for the exact conjecture, title, author, and Madden/snug terminology found the source but no later primary-source resolution or explicit duplicate of Theorem 1. The unrestricted frame-theoretic conjecture should therefore remain marked open pending expert review.

References

- [1] R. N. Ball, *Structural aspects of truncated archimedean vector lattices: simple elements, good sequences*, arXiv:1906.00439 (2019), <https://arxiv.org/abs/1906.00439>.
- [2] R. N. Ball, *Structural aspects of truncated archimedean vector lattices: good sequences, simple elements*, Comment. Math. Univ. Carolin. 62 (2021), 95–134, doi:10.14712/1213-7243.2021.009.